



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

COURSE PLAN

Department	:	Biotechnology			
Course Name & code	:	Bioreaction Engineering & BIO3153			
Semester & branch	:	5TH & Biotechnology			
Name of the faculty	:	Dr V Ramachandra Murty			
No of contact hours/week:		L	T	P	C
		3	1	0	4

Course Outcomes (COs)

At the end of this course, the student should be able to:		No. of Contact Hours	Marks
CO1:	Explain the kinetics of homogeneous reaction system	08	16
CO2:	Use the integral and differential methods for analyzing the batch reactor data	08	16
CO3:	Design single batch, continuous reactors for the given bioprocess	12	24
CO4:	Design cascade and recycle reactor system for the given process	12	24
CO5:	Analyze the RTD in continuous reactors and estimate the extent of non-ideality in commercial reactors	10	20
Total		48	100

Assessment Plan

Components	Assignments	Sessional Tests	End Semester/ Make-up Examination
Duration	20 to 30 minutes	60 minutes	180 minutes
Weightage	20 % (4 X 5 marks)	30 % (2 X 15 Marks)	50 % (1 X 50 Marks)
Typology of Questions	Understanding/ Comprehension; Application; Analysis; Synthesis; Evaluation	Knowledge/ Recall; Understanding/ Comprehension; Application	Understanding/ Comprehension; Application; Analysis; Synthesis; Evaluation
Pattern	Answer one randomly selected question from the problem sheet (Students can refer their class notes)	MCQ: 10 questions (0.5 marks) Short Answers: 5 questions (2 marks)	Answer all 5 full questions of 10 marks each. Each question may have 2 to 3 parts of 3/4/5/6/7 marks
Schedule	4, 7, 10, and 13 th week of academic calendar	Calendared activity	Calendared activity
Topics Covered	Quiz 1 (L _{x1-x2} & T _{y1-y2}) (CO x)	Test 1 (L _{a1-a2} & T _{b1-b2}) (CO x)	Comprehensive examination covering full syllabus. Students are expected to answer all questions (CO1-5)
	Quiz 2 (L _{x3-x4} & T _{y3-y4}) (CO x)		
	Quiz 3 (L _{x5-x6} & T _{y5-y6}) (CO x)	Test 2 (L _{a3-a4} & T _{b3-b4}) (CO x)	
	Quiz 4 (L _{x7-x8} & T _{y7-y8}) (CO x)		

Lesson Plan

L. No.	Topics	Course Outcome Addressed
L0	Introduction to the course	NA
L1	classification of biochemical reactions, variable affecting rate	CO1
L2	Reaction rate, kinetics of homogeneous reactions	CO1
L3	Elementary and Non-Elementary reactions, molecularity, order of the reaction	CO1
T1	TUTORIAL1-Kinetic Models, Mechanism and validation	CO1
L4	Temperature dependency of reaction rate, Interpretation of batch reactor data	CO1
L5	Searching for the kinetic model-a case study1	CO1
L6	constant volume batch reactor, integral method of analysis	CO2
T2	TUTORIAL2-Searching for the kinetic model-a case study2	CO1
L7	Irreversible bimolecular type reaction type first and second order reactions	CO2
L8	Empirical rate equation of nth order, overall order from Half-life method	CO2

L9	Irreversible reactions in parallel, Homogeneous catalyzed reactions	CO2
T3	TUTORIAL3-Numerical problems on integral method of analysis	CO2
L10	Autocatalytic reactions, Evaluation of Kinetics for Autocatalytic reaction	CO2
L11	Reversible reactions-First order	CO2
L12	Reversible reactions-2nd order with Numericals	CO2
T4	Assignment-1, TUTORIAL4- Numerical problems on integral method of analysis	CO2
L13	Reactions of shifting order	CO2
L14	Variable volume batch reactor-First order reaction	CO2
L15	Differential method of analysis	CO2
T5	TUTORIAL-5 Numerical problem on Half-life method	CO2
L16	Classification of ideal reactors and design of single ideal reactor	CO3
L17	Design of ideal batch reactor	CO3
L18	Space-time, space-velocity, Design equation of CSTR	CO3
T6	TUTORIAL6-Numerical problem on differential method of analysis	CO2
L19	Design of Ideal PFR	CO3
L20	Size comparison of single reactors, MFR vs PFR	CO3
L21	Multiple reactor system, PFR's in series	CO3
T7	TUTORIAL7-Numerical problems on Design of batch reactor	CO3
L22	Equal size MFR's in series	CO4
L23	Unequal size MFR's in series-Determination of outlet concentration from the given setup	CO4
L24	Unequal size MFR's in series-Best reactor setup for the given conversion	CO4
T8	Assignment-2, TUTORIAL8- Numerical problems on cascade reactor system	CO4
L25	Recycle Reactor	CO4
L26	Optimum recycle ratio	CO4
L27	Reactors for autocatalytic reactions	CO4
T9	TUTORIAL9-Numerical problems on cascade reactor system-design	CO4
L28	Non-ideal Reactors-RTD	CO5
L29	E, F, C-curves, Tracer input signals	CO5
L30	RTD for ideal reactors of PFR and CSTR	CO5

T10	TUTORIAL10-Numerical Problems on Recycle reactor	CO4
L31	Material balance during pulse and step input experiments during RTD	CO5
L32	Analysis of RTD data, Mean, Variance , dispersion number calculations from C-curve and E-curve	CO5
L33	RTD for cascade reactor system of PFR & CSTR, Micro and Macro mixing	CO5
T11	TUTORIAL11-Conversion calculation for Macro fluid,Numerical problems on RTD	CO5
L34	Assignment-3, Introduction to industrial bioreactors, Batch bioreactor for Enzymes and cells	CO3
L35	Enzyme catalyzed reactors, Chemostat for cell cultures	CO3
L36	Optimum dilution rate, wash out, Critical Dilution rate	CO4
T12	TUTORIAL12-Fed-batch reactor, Design equations Chemostats in series	CO4
L/T	Click or tap here to enter text.	

References:

1. Octave Levenspiel. Chemical Reaction Engineering. John Wiley & Sons, (3e), 2003
2. Harvey W. Blanch and Douglas S. Clark. Biochemical Engineering, CRC Press, 1997
3. John Villadsen. Bioreaction Engineering Principles. (3e), springer publishers 2011
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Submitted by: V RAMACHANDRA MURTY

(Signature of the faculty)

Date: 01-08-2020

Approved by: Click or tap here to enter text.

(Signature of HOD)

Date: 01-08-2020

FACULTY MEMBERS TEACHING THE COURSE (IF MULTIPLE SECTIONS EXIST):

FACULTY	SECTION	FACULTY	SECTION
